ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

Project ID: 25-26J-106

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INTEGRATING EXPLAINABLE AI AND ADAPTIVE RECOMMENDATIONS FOR HUMAN-CENTERED ROBOTIC SORTING

Project ID: 25-26J-106

Project Proposal Report

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DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

In this study, we have designed and implemented a robotic sorting system that can learn from Explainable Artificial Intelligence (XAI) for education purposes. The goal is to improve cognitive abilities, such as identifying objects, classifying objects, and making decisions, by using a learning approach assisted by technology. The project was conducted in collaboration with SERD Kelaniya, which aimed at creating an educational tool to be used in an environment that requires adaptability and interactivity.

Computer vision, machine learning, and embedded systems were used as the methodologies. The objects are identified with the use of a YOLOv8 model that was trained using an original data set that contains 2,080 images, which are annotated to identify various objects of varying colors and shapes. Grid mapping is employed to determine the locations of objects in the workspace. The recommendation algorithm uses the Random Forest, XGBoost, and Logistic Regression algorithms. The optimal sorting algorithm is determined depending on the object's distribution, and the Random Forest algorithm was chosen since it performs better and can be used for explaining the system's decision process.

The robotic execution layer consists of an Arduino-based control unit through which predefined routines allow for precise pick-and-place actions over 22 grid coordinates. Serial communication facilitates real-time exchange of data between the vision component and robot, while a stream from the IP webcam provides a flexible solution for image acquisition. Experiments yielded great success, with the YOLOv8 model demonstrating precision and recall and the recommendation model attaining 0.94 accuracy. Object sorting was successfully conducted on-the-spot, with robust robotic execution and explainable outputs.

It is concluded that combining explainable artificial intelligence and robotics can be highly beneficial in enriching the educational experience of differently-abled students.

Keywords: Explainable AI, Robotic Sorting System, YOLOv8, Random Forest, Assistive Learning

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LIST OF ABBREVATIONS

Abbreviation	Description
AAI	Adaptive Artificial Intelligence
ABB	Asea Brown Boveri (Robotics company)
AI	Artificial Intelligence
AUBO	Robotic manipulator platform (AUBO Robotics)
CBOT	Collaborative Robot
CNN	Convolutional Neural Network
DC	Direct Current
DHL	Deutsche Post DHL Group
DOF	Degrees of Freedom
DRL	Deep Reinforcement Learning
DT	Decision Tree
ESP	Embedded System Platform
FANUC	Fuji Automatic Numerical Control
GPU	Graphics Processing Unit
HRC	Human–Robot Collaboration
HRI	Human–Robot Interaction

Abbreviation	Description
HRI/P	Human–Robot Interaction / Physical Interaction
HTML	HyperText Markup Language
IP	Internet Protocol
IoT	Internet of Things
ISO	International Organization for Standardization
KPI	Key Performance Indicator
LM	Linear Model / LM2596 DC-DC converter
LiDAR	Light Detection and Ranging
LIME	Local Interpretable Model-Agnostic Explanations
ML	Machine Learning
mAP	Mean Average Precision
NLP	Natural Language Processing
PCA	Principal Component Analysis
PCA9685	PWM Servo Driver Module
PID	Proportional Integral Derivative
PWM	Pulse Width Modulation
RGB	Red Green Blue
RGB-D	Red Green Blue + Depth

Abbreviation	Description
R-CNN	Region-based Convolutional Neural Network
ReLU	Rectified Linear Unit
RF	Random Forest
RL	Reinforcement Learning
ROS	Robot Operating System
SERD	Services for Empowerment and Rehabilitation of Disabled Children and Families
SHAP	SHapley Additive exPlanations
SSD	Single Shot MultiBox Detector
SMOTE	Synthetic Minority Over-sampling Technique
TPU	Tensor Processing Unit
UAV	Unmanned Aerial Vehicle
USB	Universal Serial Bus
UX	User Experience
YOLO	You Only Look Once
YOLOv8	You Only Look Once version 8
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

1. INTRODUCTION

1.1 Background & Literature survey

1.1.1 Background

The robotic sorting system is a significant technological development that is playing a crucial role in automation today because of its capability to classify items effectively. The conventional uses of the robot include industrial, agricultural, recycling, and logistics sectors, but the latest developments have led to its incorporation even in areas such as education. As an educational tool, it can aid in the development of cognitive skills through sorting and comparisons within an educational setting, especially in cases involving disabled individuals.

However, despite significant advancements in robotics and artificial intelligence (AI), real-world deployments of sorting robots still face numerous challenges. Such problems are more likely to occur in complex environments, where the objects can be oriented in an unforeseen manner, superimpose over each other, or be in different shapes, colors, and textures. Moreover, external environmental conditions like dust, lighting differences, and noise from sensors may also affect perception and system performance.

Another significant obstacle lies in the adaptability of robotic sorting systems to dynamic and unstructured environments. In many real-world scenarios, objects arrive in unpredictable orientations, overlapping with others or partially occluded, which complicates both detection and classification tasks. Furthermore, environmental factors such as dust, moisture, or mechanical vibrations can introduce additional noise into IOT operations, reducing system reliability.

In addition, HRC adds even more complexities, especially when it comes to situations involving robot operation within an educational setting, where interactions between the machine and user become more frequent. Robotic systems must not only identify and sort objects accurately but also anticipate and adapt to changes in user workflows, safety requirements, and operational priorities. Traditional static algorithms are insufficient for these contexts, as they cannot adjust to fluctuating task demands or unexpected operator interventions. Incorporating adaptive behavior modeling, multimodal communication channels, and explainable AI frameworks can enhance trust, transparency, and overall system efficiency in such collaborative environments.

While technical performance is crucial, transparency in robotic decision-making is equally important for building trust in human-robot collaboration. Especially in education, when a robot's actions appear opaque or arbitrary, educators may hesitate to rely on it, reducing the system's overall effectiveness [9]. To address this, our system integrates Explainable AI (XAI) approaches that combine interpretable machine learning models with rule-based heuristics. The robot is capable of generating natural language explanations for its actions, supplemented by visual or voice cues to provide additional clarity. Additionally, the system supports interactive queries, enabling users to ask why certain actions were taken and receive context-aware explanations [10].

The current study involves the participation of Services for Empowerment and Rehabilitation of Disabled Children and Families (SERD), which is an institution that provides different services aimed at enhancing the lives of differently-abled children and their families through education and rehabilitation. The process of learning at SERD takes place in an interactive manner that promotes self-expression, confidence building, and cognitive skills. The integration of robotic systems in such a scenario presents a chance to develop engaging observation-based learning, where basic skills such as sorting, comparing, and recognition of objects can be learned. With this approach, Human–Robot Collaboration (HRC) can be applied in a more meaningful and impactful manner. In this

context, the use of technology extends beyond automation, serving as a tool to actively engage learners through interactive activities and alternative learning approaches.

The integration of robotic perception, decision making, and XAI allows creating an open and interactive robotic system aimed at sorting tasks in educational settings. Not only does the system automatically sort objects but helps differently-abled students to learn from the experience of observing this system's operation. Even though actions like sorting, comparing, and distinguishing between objects may seem easy for typically developing children, they can be difficult for learners who differ cognitively from others. Such actions may require more engaging and supportive approaches for such students who need help when it comes to developing the necessary cognitive skills. Thus, the goal of this paper is not only to implement new robotic systems into the process of learning but to integrate them in such a way that it can contribute to better learning outcomes. In this case, interactive robotic sorting combined with observations allows students to comprehend patterns, connections, and even grouping in an interesting way. The key feature of this paper is that unlike previous approaches which concentrated on helping average children learn better, the current one targets learners with different abilities.

1.1.2. Literature Review

1.1.2.1. Human–Robot Collaboration in object sorting

The evolution of sorting technologies has moved from purely manual processes to semi automated systems and more recently to advanced robotic systems. Sorting operations in warehouses, recycling facilities, and factories have conventionally depended heavily on human intervention due to their capacity to change and make choices. But with the move of industries worldwide towards scalability and efficiency, robot sorting systems run by automation have begun to make their presence felt, driven by robotics, machine vision, and AI. For instance, Amazon Robotics has transformed warehouse operations by dispatching mobile robots to sort and transfer goods, freeing human labor for repetitive tasks. These advancements can also be seen in logistics organizations such as DHL and recyclers such as ZenRobotics employing AI-driven robots to recognize and sort out waste streams [1], [2]. These advancements are progressing towards more intelligent and responsive robotic sorting technology with the ability to reduce cost and efficiency in dynamic settings.

Despite such advancements, the transition to shared systems from complete automation is the watermark in robotics. Collaborative robots, or cobots, are designed to coexist in workplaces safely with humans, unlike other industrial robots that must be kept at bay for safety reasons. Cobots have increasingly been utilized in assembly lines and sorting functions where human flexibility and robotic precision can complement each other [3]. Research has shown that cobots comprised approximately 5% of industrial robot sales in 2018, and the figure is expected to rise to 13% by 2022, with the global cobot market forecast to rise to nearly USD 2 billion by the year 2030 [4]. This is an indication of growing industrial recognition of HRC as a productive framework for object sorting and related manufacturing processes.

Apart from applications within the industry, current research developments also show the possibilities of HRC in contexts where the interaction between humans and machines is more geared towards education. Here, robotic devices are not just used for automation purposes; rather, they serve as means through which learning takes place. Robotic devices, through their actions, such as object sorting, which are clear and observable, can help the learner grasp basic principles like classification and comparison.

Figure 1. 1 Figure of the timeline of sorting systems

1.1.2.2. Robotic perception and manipulation

The functioning of robotic sortation systems is heavily dependent on perception and manipulation skills. Object detection methods based on deep learning are the cornerstone of robotic perception using models like You Only Look Once (YOLO), Faster R-CNN, and Single Shot MultiBox Detector (SSD) being common [5]. These models allow for real-time object localization and classification fundamental for proper robotic grasping. Moreover, modern sorting robots increasingly use visual input together with depth sensors (RGB-D) to estimate object pose, plan appropriate grasps, and generate collision-free trajectories [6]. For example, an RGB-D-based robotic system tested on AUBO

manipulators exhibited the ability to detect, locate, and grasp a range of objects autonomously, which demonstrated the power of pairing computer vision with robotic manipulation [6].

In addition to visual perception, tactile sensing is increasingly important in order to improve grasp success rates. Recent studies have emphasized visuo-haptic object perception, whereby robots integrate vision with tactile sensing to identify object properties such as texture, stiffness, and weight. This multimodal sensing approach provides more flexibility in unstructured environments where vision may not be sufficient [7]. Moreover, agile sorting technology also emerged within the recycling sector, where robots implement hybrid strategies such as push-and-drop or pick-and-release, assisted by fast trainability detection models that react differently to changing illumination and backgrounds [8]. This is a process significantly improving sorting accuracy in constantly changing conditions where items are very diverse.

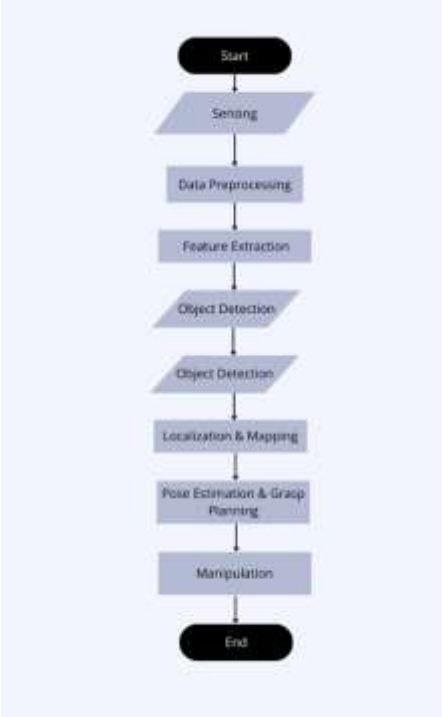

Figure 1. 2 Flow of the robot perception system

1.1.2.3. Human-Robot Collaboration (HRC)

Human–robot collaboration transcends spatial cohabitation to cognitive and decision-making integration. Cobots demonstrate this paradigm by supporting shared autonomy wherein people and robots divide tasks according to their strengths. People contribute flexibility, context understanding, and problem-solving, while robots provide speed, precision, and repeatability [3]. Adaptive collaborative control structures suggest that robots should not be passive machines but active partners capable of executing autonomous decisions through the integration of human inputs [9]. The structure is contrary to traditional master–slave control strategies and is an advancement towards collaboration at the cognitive level.

The integration of machine learning (ML) is also a factor that enhances HRC by enabling robots to anticipate human actions, learn through demonstration, and modify task execution strategies. A systematic review conducted in 2022 revealed that ML-based collaboration models increasingly emphasize time-dependent behavior, human state estimation, and environment awareness, which are crucial for industrial HRC applications [10]. Moreover, the design of the collaborative workspace has become ergonomically, inclusive, and socially created taking into account safety and comfort with the same level of efficiency [11]. Ergonomic structures have a tendency to incorporate motion planning, physical human–robot interaction (pHRI) control, and real-time safety monitoring in an effort to reduce worker fatigue and maximize cooperation [12].

Object handover is one of the most challenging but essential HRC issues. Pre-programmed handover points are utilized by most robots, which degrades fluency and increases human workload. Recent research emphasizes the need for robots to adaptively determine handover strategies as a function of human preference, object type, and situational context [13]. Similarly, transparency and trustworthiness have been identified as essential to

successful HRC. According to studies, employees resist unclear decision-making procedures, a characteristic that has been described as "algorithm aversion." Therefore, robots are likely to provide explainable feedback in order to enable humans to acknowledge the reasonableness of their decisions [14].

1.1.2.4. Human-Robot Collaboration in Education

Recent developments in Human-Robot Collaborative technology have seen a shift from its initial use in industry to include the use of robots in educational settings. In educational contexts, the aim of using robots is not to automate activities but to facilitate engagement and motivation as well as individualized learning experience for children who can learn effectively through sensory and active learning processes. Unlike other systems which simply perform tasks, educational robots serve as facilitators of learning by providing guidance during activities such as observation, interaction, and repetition, thus promoting cognitive skills like classification, comparison, and object recognition. [53], [54]

Some recent research has shown that HRC has significant effects on learners' performance if used with children who have different learning capabilities, even those who are differently abled. In this regard, SARs have been successfully implemented to develop skills related to attention, communication, and participation in tasks for children with development disabilities. The integration of perception, interaction, and adaptive feedback technologies ensures an inclusive learning environment that can adapt to learners' actions and capabilities [55]. Furthermore, the use of artificial intelligence enables robots to alter task complexity levels depending on learners' needs.

Transparency and interpretability are also important in educational human-robot cooperation (HRC), especially for teachers using these tools as pedagogical devices. The importance of explainable artificial intelligence (XAI) cannot be underestimated because it allows one to get a better understanding of decision-making mechanisms employed by robots. Numerous studies have shown that explainability is vital in enhancing the acceptance of human-robot cooperation in the classroom setting since it helps educators interpret how algorithms function and provide them with necessary feedback [56]. Consequently, the combination of human-robot cooperation with XAI and adaptive artificial intelligence (AAI) leads to the creation of intelligent pedagogical systems.

1.1.2.5. Existing Technologies, Sensors, and Hardware

In addition to theoretical and methodological human–robot collaboration advancements, the practical implementation of robotic sorting systems relies on sensing technology, robotic arms, and empowering hardware platforms. All these technologies constitute the foundation for perception, actuation, and human–robot safe collaboration.

Sensing technologies constitute the backbone of robot sorting. RGB and RGB-D cameras (e.g., Intel RealSense, Microsoft Azure Kinect) have been employed widely for object detection and pose estimation since they are capable of capturing visual and depth information [16]. Depth sensors with structured light or time-of-flight sensors allow robots to localize objects accurately in 3D space For finer control, force and torque sensors implanted within the robot wrist provide feedback on contact forces, which is necessary for delicate object handling and human-to-robot handovers [17]. In addition, tactile sensors implanted in robotic grippers enable the recognition of object surface features like texture and compliance, thereby enhancing grasp reliability [18]. Advanced research also explores multimodal sensing integration, where vision, force, and touch inputs are combined to facilitate perception of cluttered and dynamic environments.

At the hardware level, collaborative robot arms such as the Universal Robots UR series, ABB's YuMi, and FANUC CRX models have become standard platforms for HRC research. These robots are provided with joint torque sensors certified for safety and soft collision detection capabilities in order to interact with people without enclosures [19]. Gripper technologies have also leapfrogged from parallel-jaw grippers to industrial components to adaptive, anthropomorphic hands such as the Robotiq 2F-85 or Shadow Dexterous Hand, capable of dealing with diverse objects. Soft and air grippers have also been designed to handle fragile or asymmetrically shaped items and are specifically relevant in streams of waste or food products sorting [20].

Automated sorting machines also require computer hardware to support perception and decision-making. GPUs and specialized AI accelerators (e.g., NVIDIA Jetson AGX Xavier, Google Coral TPU) are normally applied onboard robots to run deep learning models in real-time [21]. Edge computing allows robots to be independent without computation involving clouds, particularly needed in dynamic industrial settings where latency must be kept low.

Finally, interaction and safety technologies play a critical role in collaborative sorting. Proximity sensors, LiDAR scanners, and 3D vision-based detection systems enable robots to detect the presence of humans and slow down or alter motion accordingly. These devices, controlled by ISO 10218 and ISO/TS 15066 safety standards, enable robots to decrease force adaptively and stop when human operators approach the workspace [22]. With the integration of perception sensors, compliant manipulators, and real-time computation, current technologies provide a basis for constructing advanced human–robot collaborative sorting systems.

1.1.2.6. Technology-Assisted Learning for Differently-Abled Children

There is considerable evidence of the immense value of integrating technology into learning, especially when it comes to children with various learning challenges. Interactive and engaging techniques have been found to be highly efficient in facilitating cognitive and motor skill development among differently-abled children. Traditional learning approaches may fail to yield the desired results when it comes to topics that need visualization, repetition, and interactive practice, such as comparing objects and sorting activities.

Organizations such as SERD adopt an inclusive approach to learning where children are encouraged to interact and express themselves confidently. In such an environment, the integration of robotic technology can enhance the process of learning through observing, interacting, and learning from feedback. The robot can be employed as an interactive tool that showcases the task repeatedly in a consistent manner to encourage students to observe and replicate the actions while staying focused.

In this study, the design of the sorting robotic system that has been proposed is meant to facilitate the achievement of these learning goals through the provision of an opportunity for the learner to observe and engage in a systematic sorting activity according to specific characteristics of the object being sorted such as color or shape. In addition, the robot's influence does not end there as it facilitates the involvement of both the teacher and learner in collaborative learning. Through recommendation-driven decision-making and explainable AI, the system sorts and educates at the same time.

1.1.2.7. Challenges in current systems

Despite the impressive developments in robot sorting systems, several limitations still prevail. To begin with, adaptability to changing environments poses a serious challenge. Real-world sorting is often characterized by diverse object geometries, occlusions, and environmental conditions. Although deep learning algorithms are extremely accurate in regularized environments, their performance is compromised in dynamic or new situations unless they are retrained. Agile methods such as quick model adaptation and human-corrected correction have been proposed to address this issue [8]. Indeed, experiments show that human interaction, i.e., occlusion removal or grip pose adaptation, can increase sorting accuracy from ~33% to ~100%, showing the benefit of hybrid collaboration [15]. In the field of education, this flexibility is all the more important, since dealing with students adds another layer of uncertainty.

Secondly, inexplicability of robotic decision-making demoralizes trust and transparency. The majority of sorting robots are "black boxes," where human beings can't comprehend why specific objects were sorted or treated in specific way. Explainable AI (XAI) frameworks are therefore required to make the decision-making process clear so that human workers can build up trust and work effectively alongside robots [14]. When it comes to educational institutions, transparency becomes even more vital since it would allow educators to comprehend the way the system operates and teach students through its means.

Thirdly, optimization of efficiency still remains challenging where there are multiple kinds of objects. Bots find it hard to optimize the prioritization of objects or acquire task strategies in complex situations. Furthermore, methods based on data such as reinforcement learning, as promising as they are, have limitations such as sample inefficiency, sparse rewards, and sim-to-real gap [13]. Safety and ergonomics concerns

also remain: cobots tend to be slower than traditional robots due to safety constraints, which could impair throughput if addressed through sophisticated control and scheduling algorithms [11]. These limitations play an important role within the academic setting, although they could influence the responsiveness and interaction capabilities of the robots if not adequately addressed. Thus, solving these issues is vital when creating effective and functional robots which can be used both in industries and academia.

1.1.2.8. Explainable Artificial Intelligence (XAI) in Robotics

1.1.2.8.1. Introduction to XAI

Explainable Artificial Intelligence (XAI) has been a central research focus in recent years, especially since artificial intelligence systems are being employed to support decision-making for safety-critical domains such as healthcare, finance, and robotics. While traditional AI models—and deep learning architectures especially—provide state-of-the-art performance for object detection, classification, and decision-making, they behave like "black boxes" whose rationale is not known to the end users. This lack of transparency is especially troublesome when robots are being deployed in environments in which they will need to cooperate with humans, since humans will struggle with robotic decisions being difficult to understand, anticipate, or trust [23]. Seeking to resolve these challenges, the Defense Advanced Research Projects Agency (DARPA) established the XAI program in 2016, in which explainability was explicitly designed as the ability of a model to facilitate its decision-making processes to be understandable by humans [24].

In general, explainability methods can be categorized into intrinsic approaches—where interpretability is built into the model design itself (e.g., linear models, decision trees, symbolic reasoning)—and post-hoc approaches, where explanations are generated after a

model has already produced predictions, i.e., visual heatmaps or feature attributions [25]. Intrinsic methods are generally simpler but limited for addressing complex tasks like robot vision, while post-hoc methods allow complex deep learning models (e.g., convolutional neural networks) to be partially interpretable without sacrificing accuracy. For robotics, both kinds are essential: intrinsic ones help in transparent task planning, and post-hoc ones interpret perception and recognition tasks.

1.1.2.8.2. XAI Methods for Perception Tasks

One of the most extensively researched uses of XAI in robotics is visual perception operations like object detection and classification, which form the foundation of robotic sorting and manipulation. These have been successfully employed with more recent studies, such as perturbing input features to find their contribution towards predictions with Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP), which utilizes importance scores over each input variable derived through cooperative game theory [26]. For image-based robotics vision tasks, techniques like Gradient-weighted Class Activation Mapping (Grad-CAM) have proved to be extremely helpful as they create heatmaps throughout an image to show which areas affected the model's classification the most [27].

In sorting robots, for example, Grad-CAM can highlight the specific edges, textures, or colors that resulted in the vision system classifying an object as a recyclable plastic bottle instead of metal scrap. This not only helps engineers debug model actions but also helps operators understand why the robot acted in a certain way, increasing collaboration. Evidence from post-2021 research has demonstrated the effectiveness of combining SHAP with deep vision models such as YOLOv7 and EfficientDet for real-time interpretability in robotic perception systems [28]. These methods ensure that

explanations remain in accordance with human-identifiable visual aspects, thereby bridging the gap between robotic reasoning and human understanding. Recent studies demonstrate the effectiveness of Grad-CAM in robotic manipulation tasks, showing that such explanations not only help developers debug models but also allow human collaborators to validate robot decisions in real time [29].

1.1.2.8.3. XAI in Decision-Making

Aside from perception, robots need to make higher-level decisions about task accomplishment, order, and collaboration. Traditional interpretable models such as decision trees and rule-based systems provide explainable mechanisms for reasoning but are not generally flexible enough to deal with adaptive environments [30]. More current research integrates reinforcement learning with symbolic reasoning or heuristic rules in an attempt to find a compromise between adaptability and interpretability [31]. For example, warehouse robots in logistics have been equipped with hybrid AI systems where deep reinforcement learning governs object assignment and decision trees provide interpretable explanations to operators [32]. The hybrid allows tracing back the decision, such as why a robot placed a fragile object on top of another, for purposes of accountability and efficiency.

For this particular study, an analogous method would involve the use of a decision model guided by recommendations and Explainable Artificial Intelligence (XAI). In this case, the decision on priority for sorting would be made according to the features of the objects detected (for instance, their shape and color) in real-time, while the XAI algorithms such as SHAP would help understand the decisions and factors behind them. It is especially useful in education when teachers need an explanation for how the robot operates in order to assist students' learning process better.

1.1.2.8.4. XAI in Human-Robot Interaction (HRI)

In Human-Robot Collaboration (HRC), explainability has a direct consequence on trust, safety, and collaboration. Studies show that robots providing explanations of the task they perform result in higher human trust and improved task coordination [33]. In healthcare, treatment-recommending assistive robots that explain increase patient compliance and caregiver confidence [34]. Autonomous vehicles also included natural language explanations of route decisions so that passengers can understand safety-driven maneuvering [35]. In warehouse robotics, robots in industry increasingly apply XAI to explain why they select particular objects for sorting, reducing the contribution of human error and the potential for disturbances in the workflow [36]. These applications demonstrate not only how XAI enhances technical robustness but also how it supports the social dimension of HRI, enabling more seamless human-robot integration into human-centered spaces.

1.1.2.8.5. Limitations of Current XAI Approaches

While there have been developments in recent times, the current generation of XAI techniques faces some difficulties when used in robotics applications, especially when it comes to decision-making and interactive settings. The first problem encountered is that a lot of the currently available XAI methods are mostly tailored to static tasks such as image classification, and hence are not easily applicable in robotics where decision-making is needed [37]. As an illustration, although techniques such as Grad-CAM are useful in helping users understand why an image has been classified in a certain way, they fail to provide information on why certain actions were chosen in performing that classification. The second limitation is that most of the current XAI techniques require a

lot of computing power and cannot be easily used in robotics systems requiring real-time responses [38].

Finally, less work has been done in exploring the use of XAI for task-oriented and optimization-based robot design. Few of these studies consider issues regarding the explainability of the robot's task prioritization, resource allocation, and generation of actions from several sources of input [39]. This issue is especially important when the robot design process is applied in an educational setting since educators will need comprehensible explanations in order to make decisions on the robot system.

1.1.2.9. Recommendation Systems for Efficiency Enhancement

1.1.2.9.1. Overview of Recommendation Systems

Robotics recommendation systems are decision support mechanisms that provide best action prescriptions with the information at hand. Unlike traditional recommendation systems for personalization, in robotics they are also optimized for enhancing workflow performance. For object sorting tasks, it entails finding the optimal sequence of actions to minimize time consumed and maximize throughput. More recent work has married machine learning algorithms with sensor data to create learning and self-adaptive recommendation systems [40].

1.1.2.9.2. Sorting Applications

In the industrial setting, recommendation systems have found application in various tasks to improve productivity. In recycling plants, robots utilize sensor data to choose the best order of sorting, for instance, processing plastics first before metals or glass [41]. Also, in

warehouses, robots examine package characteristics such as shape and size and decide on the best handling order to reduce motion complexity and improve throughput [42]. In manufacturing, task assignment becomes optimized by recommending the sequence in which components are combined, optimizing the production process [43]. Such uses demonstrate the potential of recommendation systems to enhance productivity in applications such as robotic object sorting tasks.

In the case of the current research work, a similar recommendation-driven system is used in the field of education through a robot. Instead of concentrating on the operational efficiency, the sorting process is made dependent on the properties of the objects such as color and shape. In one example of the process, the robot sorts the objects on the basis of the maximum number of colors and shapes that appear. The method not only increases the efficiency of the operation by minimizing redundant movements performed by the robot, but also adds to the learning process by aiding students in grasping the idea of grouping, comparing, and finding patterns. Thus, recommendation systems are not limited to industry purposes but can be extended to other fields such as education.

1.1.2.9.3. Real-time Decision Support in Robotics

Real-time decision support in robotics heavily relies on sensor inputs to perceive the environment.

- RGB and Industrial Vision Cameras: These sensors detect visual characteristics such as color and shape, which help in object classification.
- Depth Sensors (e.g., Intel RealSense, LiDAR, Stereo Cameras): These sensors provide 3D data, which helps in detecting object sizes and spatial orientation.

- Spectral/Infrared Sensors: These sort by reflectance properties, useful for distinguishing between metals and plastics.
- Force-Torque or Weight Sensors: These detect material weight, useful in sorting heavy versus light objects.
- Proximity and Ultrasonic Sensors: These detect object presence and relative location, useful for path planning and collision avoidance.

By integrating data from such sensors, robots can calculate object counts per category (material, shape, size), as the basis for recommendations like prioritizing the sorting of plastic balls over others due to their abundance or postponing sorting metal boxes due to their weight and smaller quantities [44]

1.1.2.9.4. Task Scheduling and Prioritization in Robotics

Robotics task scheduling is strictly an advising task—choosing the best sequence of actions. There are various techniques employed in robotic sorting:

- Heuristics: Low-level regulations like putting the biggest type first or handling fragile items (e.g., glass) first to prevent breakage risk.
- Spatial Reasoning: Having higher priority for closer objects to minimize travel distance.
- Time-based Logic: If there is a moving conveyor belt, having higher priority for products that risk getting pushed out of the workspace.

These scheduling recommendations optimize efficiency and reduce task completion time [45].

1.1.2.9.5. Rule-based + ML Hybrid Recommendation Approaches

Hybridizing machine learning (ML) with rule-based systems provides a balance approach to recommendation in robotics. Rule-based systems are easy to understand and excel at handling well-known cases. In contrast, adaptive recommendations using ML can learn best strategies from task history. A balance approach leverages the strengths of both: ML handles object detection and feature extraction, while rule-based heuristics provides recommendations as interpretable and explainable to human collaborators [46].

1.1.2.9.6. Robotic & Automation Recommendations

Recommendation technologies increasingly find application in various robotic and automation use cases:

- Multi-Robot Systems: Recommendations decide which robot handles which set of objects to achieve optimal task allocation and efficiency.
- Sorting Lines: Optimizing through prioritization can reduce bottlenecks by separating similar items, for maximum throughput.
- Material Handling Robots: Sensors combined with rules facilitate efficient sequencing, for instance, removing fragile glass first to avoid accidents.
- Warehouse Robots: Sizing/shaping items in order of priority for stability when stacking to ensure safer and more efficient operations.

These examples establish the context of how recommendation logic can be directly transferred to robotic object sorting tasks [47].

In this research work, however, recommendations are made within the context of an education robotic application in which decisions are based on the properties of the objects, such as color and shape. While traditional systems only emphasize industrial applications for efficiency purposes, our robotic application incorporates the use of a recommendation-based method (such as color dominance) in order to achieve efficiency and maximize learning. The student can be able to identify certain patterns in the sorting process, while at the same time helping the robot avoid unnecessary motions.

Despite the advancement, there are still literary gaps. Concern for Task Allocation and Route Planning is one. Existing recommender systems for robotics are primarily focused on task allocation and route planning, whereas there is limited concern for real-time efficiency-oriented recommendations for object sorting.

Then insufficient Real-time Implementations. Few implementations of real-time recommendation systems for dynamic robotic tasks have been made, and this has been identified as an area in need of more research. Insufficient Incorporation of Explainability is another major issue. Almost no studies explain why a recommendation is being produced (Explainable AI), which is crucial in human-robot collaboration. This may be addressed in making more efficient and better-reliability robotic systems [48].

1.2. Research Gap

Recent advancements in Explainable Artificial Intelligence (XAI) have helped a lot in providing transparency, interpretability, and trust in AI systems across many domains. Adadi [49] presented a unified XAI framework for healthcare and outlined the importance of interpretable and reliable decision support for clinicians. While this framework highlights the potential of XAI for critical applications, its methodology was initially designed for static datasets and not real-time decision-making, as required in dynamic settings such as human-robot collaboration.

In finance, Wang et al. [50] investigated the accuracy-interpretable trade-off, highlighting how XAI can improve decision-making for financial prediction complex tasks. Their research was capable of successfully illustrating model explanations to end-users but was primarily limited to tabular data, and issues with multimodal data—visual and sensor inputs—were largely left untackled. Rossi and Nguyen [51] also introduced a hybrid that combined model-agnostic and model-specific XAI techniques in time-series forecasting. Although effective in characterizing temporal patterns, the algorithm struggled to scale, particularly with high-dimensional real-world data for large-scale problems, which restricts its application in robotics where rich environments in terms of sensors provide rich data streams.

In addition, Kumar et al. [52] studied XAI for natural language processing to enhance human-AI collaboration by offering interpretable text explanations for model predictions. Although this research demonstrated the value of human-centered explanations, its usage was limited to text data and did not consider multimodal explanations that include visual, audio, or interactive signals—features crucial for building trust in real-world robotic systems.

While these studies collectively advance the theoretical and applied understanding of XAI, they reveal a number of gaps regarding human-robot collaboration. First, current XAI methods are largely domain-specific and have not been tailored for real-time robotic decision-making in which the action must be explained while it is being taken. Second, multimodal explanations that combine visual, auditory, and textual feedback remain underexplored and fail to hinder effective human understanding and trust. Third, existing XAI approaches fail to provide scalability and dynamic flexibility, which are crucial in robotic sorting activities that must operate in uncertain conditions.

This issue becomes even more apparent in educational settings that use robots to interact with children with disabilities, where adaptability and explainability become critical features. Robots currently available have mostly been developed for efficient use in manufacturing and not for education through observation, interactive teaching, and feedback. There has been no effective inclusion of XAI approaches that provide important and educator-oriented information on how decisions are made by robots.

In order to solve the issues raised, this paper suggests using a real-time robotic sorting application that combines deep learning approaches to detect objects with a recommendation approach to decision making and Explainable AI algorithms. In this case, the sorting strategies of the robotic application are formed with the help of the machine learning models according to the features of the objects, and SHAP-based explanations are used to make recommendations understandable for educators.

Table 1.1: Table of identified research Comparison

Features / Gaps	Adadi [1]	Wang et al. [2]	Rossi & Nguyen [3]	Kumar et al. [4]	Proposed Research
Real-time decision support	✗	✗	✗	✗	✓
Human-robot collaboration	✗	✗	✗	✗	✓
Multimodal explanations (visual/audio/text)	✗	✗	✗	✗	✓
Adaptive strategy recommendations	✗	✗	✗	✗	✓
Interactive natural language queries	✗	✗	✗	✗	✓
Rule-based + interpretable ML integration	✓	✓	✓	✗	✓
Scalability to dynamic / large datasets	✗	✗	✗	✗	✓
Domain-specific adaptability	✓	✓	✗	✓	✓

1.3. Research Problem

Robotic systems require reliability, flexibility, and interpretability. For instance, in the case of robotic object sorting, various parameters like the nature of the objects to be sorted, shape, color, or location may need to be considered at the same time. The use of machine learning techniques can achieve the desired objective since robotic systems follow predetermined algorithms. However, the lack of transparency in the decision-making process leads to a mistrust relationship between human users (like teachers) and robotic systems because decisions by such robots become hard to comprehend. In an educational environment, it is essential to have a teacher who can understand and direct the robot during the activity.

The latest developments in Explainable Artificial Intelligence (XAI) have shown promising results in increasing the interpretability of models within applications from healthcare, finance, to natural language processing. Such techniques improve the level of understanding of the models and trust in their decision-making process through providing the necessary explanations. Unfortunately, most of the currently available explainability tools are developed with static environments in mind and cannot be used for real-time robotics. In addition, most approaches to explainability are mostly focused on text explanations and lack support for multimodal interaction interfaces.

In addition, current robotic recommendation systems concentrate on allocating tasks, planning paths, or optimizing efficiency within industrial applications. Such systems lack the capacity to provide timely and understandable feedback as to why certain decisions or actions were made. In addition, most contemporary algorithms lack dynamic adaptation to new situations or interactive learning scenarios. Therefore, there are limited

applications of such recommendation systems in education, especially in scenarios that involve students with varying abilities.

Thus, the research problem that will be investigated in this project is to develop an intelligent robotic sorting system that combines object recognition, recommendation-based decision-making, and explanations for educational purposes. In addition to efficiently sorting objects using their features like colors and shapes, the system should also be able to generate comprehensible explanations about its decision-making process. In this regard, the proposed system will utilize recommendations from machine learning with Explainable Artificial Intelligence techniques to improve transparency and trust among users. Addressing the outlined research problem will ensure that the gap between intelligent robotic automation and educational support systems can be bridged.

1.4. Research Objectives

1.4.1. Main Objective

The primary goal of this study is to design an XAI-driven robotic object sorter system that will improve learning and interaction within the educational setting among the specially abled children. The proposed system consists of an object detection model based on YOLOv8 and a recommendation model based on machine learning to perform a color or shape dominated sorting task (red, blue, green) with use of various objects including ball, box, and pyramid. To develop recommendations, a random forest-based decision module is used whereas, for interpretability purposes, SHAP-based explainable artificial intelligence system has been considered in this case.

1.4.2. Specific Objectives

Out of the research gaps in Explainable AI (XAI) for robotics and real-time recommendation systems, the specific research goals of this research are:

- **Develop an Image-Based Training Framework**
  - Build a dataset of object sorting scenarios (e.g., 5,000+ images with varying shapes, colors, and counts).
  - Annotate the dataset with both sorting order and reasoning factors (count, distance, reachability).
  - Train a model capable of learning adaptive sorting strategies from these examples.
- **Implement Object Recognition and Counting**
  - Enable the system to detect and classify predefined objects (e.g., three shapes × three colors) within an image.
  - Provide automated identification of object counts and positions to support real-time recommendations.

- **Design a Recommendation-Based Sorting Strategy**
  - Compare multiple machine learning models for decision-making.
  - Select one as the final model
  - Implement a mode-based sorting strategy to prioritize objects based on frequency.

- **Robot sorting integration in a real-time system**
    - Mapping detected objects into predefined grids for robot execution.
    - Enable Arduino robot arm control for sorting in color bins defined by the machine learning model.
    - Execution of sorting tasks one after another per recommendation by the machine learning model.

- **Creation of an Explainable AI Model**
    - Employ SHAP for explainability in machine learning model prediction.
    - Explain the decision to teachers visually as well as in a textual format to improve trust.
    - Interpretability in human–robotic interaction in educational environments.

- **Facilitate Learning Experiences via Robot-Assisted Tasks**
    - Allow teachers to lead students in multiple robot-assisted sorting tasks.
    - Facilitating observational learning for kids wherein students learn to sort, compare, and classify objects with the help of robots.
    - Assess the effectiveness of robotic assistance in engaging differently-abled kids in learning tasks.

- **System Performance Evaluation and Usability Testing**

    - YOLOv8 detection efficiency and machine learning classifier evaluation.
    - Test efficiency of the system during live sorting operations.
    - Interpretability testing and user trust in the Explainable AI feature using educator feedback.

3. METHODOLOGY

3.1. Methodology

The methodology applied in this research is a system-based approach to developing and implementing an Explainable AI-enabled predictive robotic sorting system. It combines computer vision, machine learning, Explainable AI, and robotic action into one pipeline working in real-time. In contrast to systems that are based only on automation, the proposed work places emphasis on explainability, flexibility, and educational capabilities within a teacher-student environment.

3.1.1. System Design and Workflow

As mentioned above, the proposed solution can be described as a pipeline connecting the stages of perception, reasoning, explanation, and execution. Firstly, an incoming video feed of objects located on the working surface should be provided. Objects need to be detected and categorized using a deep learning algorithm; then, the spatial locations of those objects will be projected to a predefined grid. After that, a predictive algorithm will suggest the best strategy for sorting those objects, and such strategy will be explained via XAI and executed by a robotic arm.

Overall, the solution works in an iterative way, which means that the system will constantly update its perception of the environment.

3.1.2. Data Acquisition and Object Detection

For both perception and decision-making to take place, a customized dataset has been generated in diverse but controlled environments.

The dataset consisted of a total of 2080 pictures with varying arrangements of objects on a horizontal plane.

Figure 3. 1: Sample images from the dataset

Figure 3.1 shows the sample of images in the dataset. The variations included:

- Number of objects
- Arrangement
- Overlap and orientation
- Lighting conditions
- Different surfaces
- The device used to capture the photos
- The distance and angle at which the photos were captured

All the images were annotated using Roboflow, and each object in an image belonged to any one of the nine classes, depending on their color and shape:

- blue_box
- blue_pyramid
- blue_ball
- green_box
- green_pyramid
- green_ball
- red_box
- red_pyramid
- red_ball

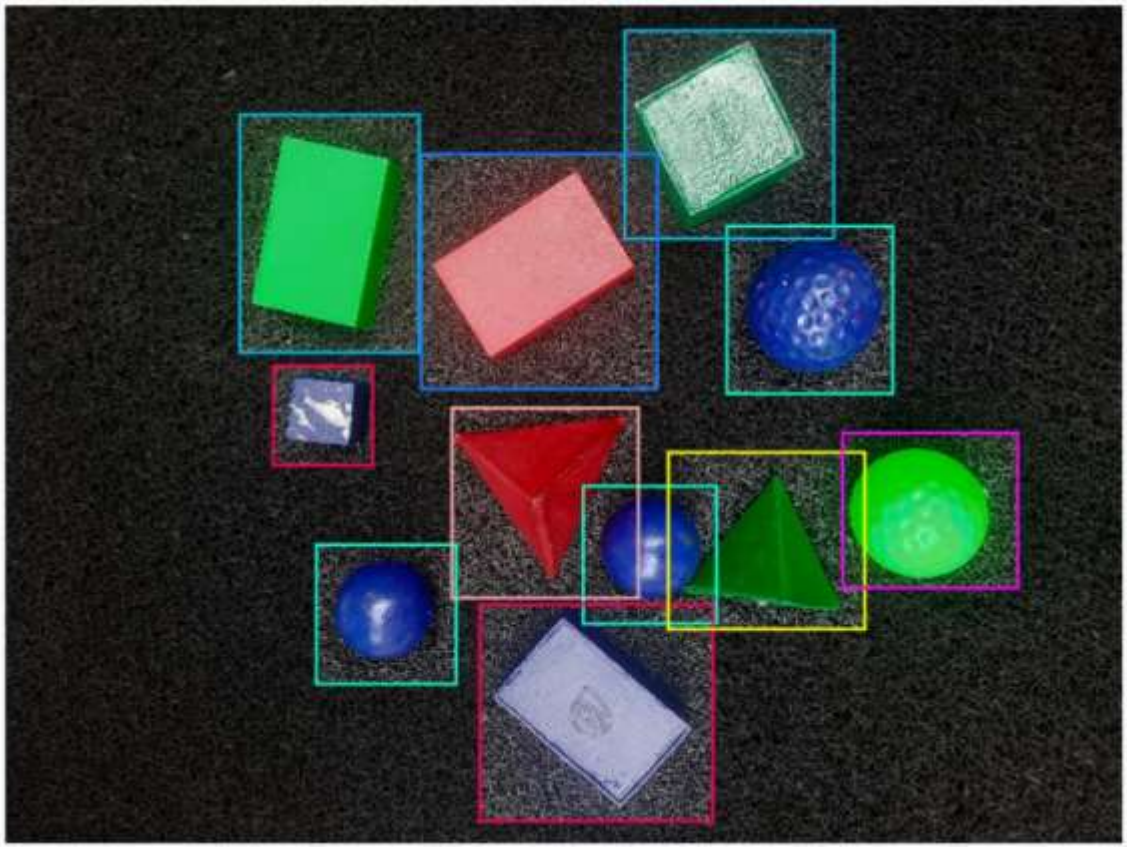

Figure 3. 2: Image after annotation

This classification mechanism helps the system recognize both objects and group them semantically for reasoning during the recommendation phase.

3.1.3. Spatial Mapping and Scene Understanding

Object detection alone is not enough for robotic interactions; it needs to be accompanied by the recognition of object positions in terms of real-world space.

For this, initially, the working surface itself (board) is recognized via edge and contour detection, after which homography transformation is used to project the camera perspective into the normalized world space.

Further, the board is divided into 22 regions, which correspond to possible positions where objects are available for interaction. Each object, after being transformed into world coordinates, is associated with one of these positions.

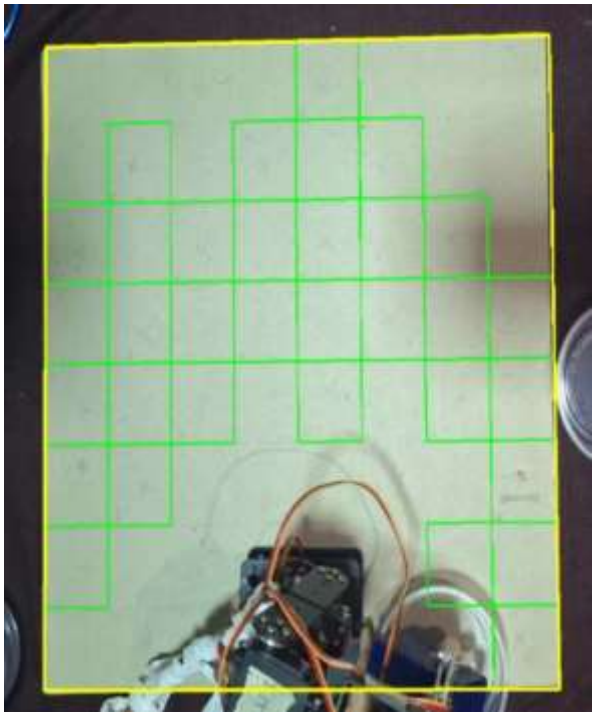

Figure 3. 3: Figure of mapping

Thus, we can conclude that this method allows us to:

- Localize objects precisely.
- Make the process robust with respect to different camera angles.
- Maintain spatial reasoning invariance between various scenes.

3.1.4. Model Development and Evaluation

One of the significant contributions made by this study is the creation of a recommendation system for predictions, which helps in determining the priority in sorting objects. The system uses machine learning methods to make decisions based on data, rather than using hard-coded rules alone.

3.1.4.1. Exploratory Data Analysis and Preprocessing

Prior to building the predictive models, the dataset went through an exploration stage to make sure it is of high quality and consistent.

The initial analysis was done on the data set to determine the distribution of object counts, sorting strategy, and imbalance classes. As discovered, there was an imbalance in some sorting strategy classes that might lead to bias in predictions by the model. The imbalance in the data set was corrected using approaches like SMOTE (Synthetic Minority Over-sampling Technique).

Features engineering involved converting the raw observation to structured inputs, such as:

- Number of objects of each type (color x shape)
- Aggregate features, including total number of objects by color
- Mode lesson indicator (one-hot encoded)

Additionally, categorical features were transformed to numbers, and for some models, feature scaling was done to scale the features. Finally, the dataset was split in a train-test ratio of 80-20.

3.1.4.2. Random Forest Model

The main approach applied in this paper involves a Random Forest model that aggregates several decision trees to generate reliable and understandable forecasts.

Figure 3. 4: Dataset used for all models

In this case, the Random Forest classifier was enhanced with multi-output classification, which makes it possible to forecast several sorting targets at once. The model utilizes decision trees trained on various partitions of input data, and their predictions are aggregated to generate the final outcome.

The parameters of the model include:

- 200 decision trees (estimators)
- Depth limitation to avoid overfitting

One of the significant benefits of using the model is its ability to compute feature importances, which helps to meet the explainability requirement. For instance, the model can specify whether the number of objects or lesson mode was more significant for the decision.

At the evaluation stage, the model proved to be effective in modeling non-linear dependencies between object distributions and sorting algorithms.

3.1.4.3. XGBoost Model

To investigate further potential for improving the performance, a model based on XGBoost was developed. XGBoost, unlike Random Forest, grows its trees one after another, where every next tree learns and compensates for the mistakes made by previous models. In this way, it becomes possible for the algorithm to discover more complex patterns.

Important characteristics of this model development are:

- Multiclass classification problem settings
- Selection of hyperparameters such as learning rate, maximum depth, and estimators

Despite being highly accurate in predictions, XGBoost model is computationally heavier and less explainable than Random Forest.

3.1.4.4. Logistic Regression Model

Logistic Regression is a simple but effective algorithm that can be used as the base model.

In this model, there is a linear relation between inputs and the probability of belonging to a particular class. To make sure that the model performs well, the inputs were scaled. Even though Logistic Regression is simpler to understand and interpret, it performed poorly when it came to capturing more complicated relations between the features.

3.1.5. Explainable AI (XAI) Integration

The XAI module becomes the critical part of this study, which turns this algorithm into an interpretable decision-making system. Although the prediction module identifies the action to be taken, the XAI module takes the responsibility to provide reasons behind the recommended action so that the system becomes transparent, reliable, and user-friendly – even within the educational environment of this study.

This system does not regard explainability as an afterthought, but it is embedded into the decision-making process. The reasons provided by this system depend on the same factors on which the recommendation depends, such as object count, colors, groupings, etc. Therefore, the explanations do not become approximations of the true reasoning; instead, they reflect the decision-making process accurately.

3.1.5.1. Model-Based Explainability Approach

For the explainable AI part of the system, the idea behind it is to have an understanding of the reasoning behind the predictions made by the sorting algorithm without any change in its efficiency. The use of explainability in this case is realized through model-based explanation, which means that the explanations will be provided based on the actual algorithm rather than on surrogate algorithms. Since the recommendation engine works with a set of structural features such as counts of objects with different colors and shapes, together with the sorting mode selected, SHAP is used.

SHAP method will be quite useful in the current situation since it can generate consistent and well-supported explanations for any decision using the cooperative game theory principles to establish the importance of each individual variable. Unlike the predictive model where the output is treated as a "black box," SHAP looks into the prediction and sees the effect of each combination of variables on the outcome. As a result, the system can find out which variables such as the prevalence of a certain color or the shape play the most important role in determining the optimal strategy of sorting the objects.

3.1.5.2. Explanation Generation Mechanism

Following the computation of SHAP values during the inference phase, these values are used to derive information that is understandable enough to present to the user. This involves ranking the contributions of each feature, and then deriving logical interpretations based on such information. For example, if there is a certain color that contributes the most positively to the recommended strategy, the system will interpret this feature to be responsible for the recommendation.

Instead of providing plain numerical representations of SHAP values, which would be quite hard to understand, the system provides a way to translate these values into an explanation. As a result, these explanations are dynamically created depending on the current scene and the output from the model. In effect, the process of creating explanations can be viewed as a process of translating reasoning done by the model.

3.1.5.3. Visual and Interactive Explainability

In order to increase interpretability even further, visual explanation methods have been included in addition to text explanations within the system. For example, in the main interface, there are graphs that show the distribution of identified objects according to color and shape. These elements correspond to the features considered by the algorithm, thus allowing the user to verify the reasoning behind each suggestion.

Furthermore, the chosen option is marked, and a detailed explanation of the process of reaching the conclusion is provided. Through the use of text and graph explanations, users can not only analyze data but also evaluate how the decision was reached. The design of the interface takes into account the fact that it will be utilized in the field of education and that the users might not possess any technical knowledge.

The interactive aspect of the interface also provides the ability for the user to see how the explanation will vary according to the changing object configuration. This will further strengthen the association between the inputs and the output produced by the system.

3.1.5.4. Integration with Real-Time Decision Making

One of the important prerequisites of this approach is the generation of explanations on the spot when the robot makes actions. This way, the XAI component is deeply intertwined with the perception and recommendations pipeline. After completing the object detection phase and freezing the state of the system for further analysis, both recommendations and their explanations are generated simultaneously.

The immediate availability of such explanations is crucial for making sure that there will not be any inconsistency due to the dynamic changes of the system. In this regard, the explanation will always coincide with the moment of the recommendation's generation, ensuring consistency with the initial reasoning conditions. The possibility to generate explanations in real time is vital for establishing user trust in the system.

3.1.5.5. Role in Educational Context

In addition to the transparency provided by the XAI element, there is a potential for the improvement of educational significance of the whole system. Specifically, the explanations have been prepared in compliance with basic learning principles like counting, grouping, and comparison. As a result, students will be able to understand the reason behind the robot’s actions on their own and learn through imitation.

There is a possibility for teachers to employ explanations as a teaching method; they can instruct their pupils to watch the system and reproduce its reasoning manually afterward.

At the same time, consistent reasoning and explanations guarantee that learners comprehend the basics instead of just learning conclusions.

Finally, it should be mentioned that the evaluation of students' actions according to the principles used by the system is a way to make a full circle from machine to learner since the robot evaluates sorting performed by the learner.

3.1.6. Robotic Control and Execution

The robotic actuation layer is tasked with the physical implementation of the output from the vision and recommendation layer. The layer achieves this via the use of an intelligent integration of software and hardware systems. The design guarantees that the results from object detection, sorting, and movement are implemented in a timely manner, with stability and accuracy when performing robotic manipulation tasks.

3.1.6.1. Hardware Components

The robot design utilizes numerous hardware units working together for movement, power, and actuator controls. The central processing unit of the design is the Arduino MEGA 2560, a microcontroller board with an integrated circuit that serves as the primary device for accepting instructions from the vision control and performing programmed movements. It has ample inputs and outputs to accommodate several actuators at once.

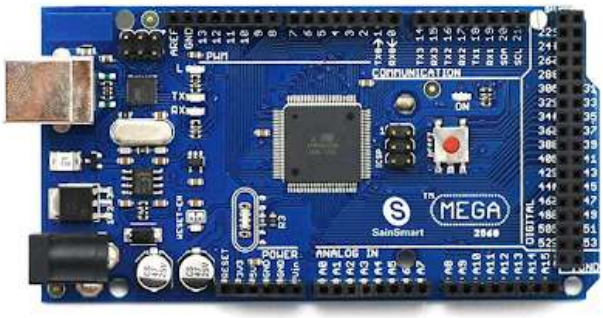

Figure 3. 5: Arduino MEGA 2560 board

The PCA9685 PWM driver is the module used for controlling the servos, a 16-channel, 12-bit pulse-width modulation controller designed specifically for controlling many servo motors at the same time. This component is vital because it relieves the servo timing from the processor, resulting in better multi-joint manipulation of the robotic arm.

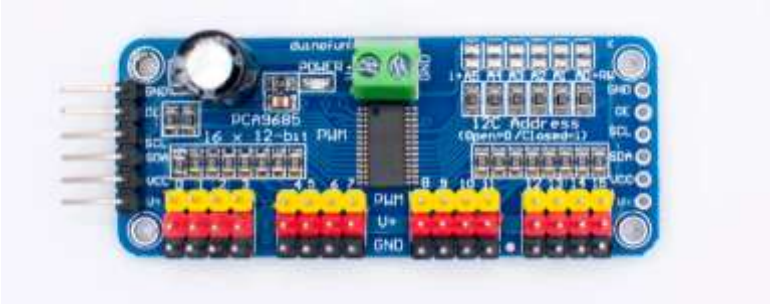

Figure 3. 6: PCA9685 PWM driver

Stable power will be delivered using the LM2596 DC-DC buck converter that will step down the 12V source to 5V as required by the servo motors and the control electronics. This is important since the servo motors are quite susceptible to voltage fluctuations. Stable power is therefore essential in order to prevent the jittering of the servo motors and any damage that may arise.

Figure 3. 7: LM2596 DC-DC buck converter

There are many servo motors in this project. The servo motors act as actuators and will each be tasked with controlling the different joints of the robotic arm like the base rotation joint, shoulder joint, elbow joint, wrist joint, and gripper joint.

Figure 3. 8: Micro Servo MG995

Lastly, there will be the 6-DOF 3D Rotating Mechanical Robot Arm which serves as the manipulation unit of the system.

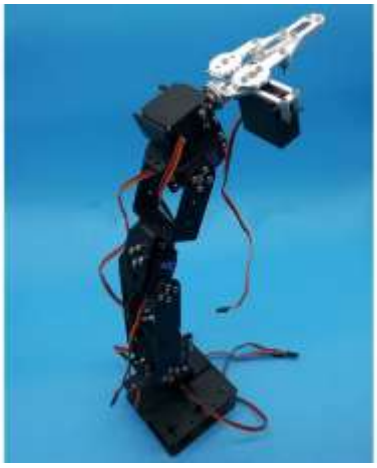

Figure 3. 9: 6-DOF 3D Rotating Mechanical Robot Arm

3.1.6.2. Arduino Control Logic

The motion plan is defined by the code written within the Arduino layer, indicating the method of performing the motion. For each identified object, there exists an associated function representing the action that should be done. Some example functions are pickBox1, pickBox2, etc., and these correspond to instructions on how to move the robotic arm using a sequence of angles.

Since sudden movements will put strain on the mechanical joints and possibly cause mechanical failure, a smooth motion algorithm is employed. This ensures that the movement is gradual and that the servos move in small steps until reaching the required angle. Moreover, the motion of the gripper part of the robotic arm can happen separately.

A typical workflow for the execution phase in Arduino involves several distinct stages. The first stage requires positioning the robotic arm at the default starting position to be able to safely proceed to the next steps. The robot then goes to the position of the found object's coordinates. After proper alignment, the joint movement is performed to obtain

the best angle for gripping the object. The gripper clamps and picks up the item after that, and the robotic arm moves to the position where the particular object should be placed according to the color and shape classification mode.

3.1.6.3. Communication and System Integration

The interaction between the vision system-based intelligence and the robotic hardware is achieved through serial communication via USB, allowing real-time sending of commands from the Python program to the Arduino microcontroller. This is done through the use of structured command inputs in the following format:

- box_id,object_type
- Examples include: 1,blue

In this manner, the Arduino is able to understand not only the position of the object but also its category. Upon processing the command, the Arduino initiates the appropriate motion routine of the robot toward the designated box and places the object accordingly into the appropriate bin (red, blue, or green).

In terms of the perceptual component, the system utilizes a stream from an IP webcam. In particular, in our project, a smartphone camera will be the main sensory element. The video stream will be taken from the network by means of a URL in real time. Thus, it gets rid of the need for expensive industrial cameras or ESP32-CAM modules. Also, improve accessibility, as there is no need for teachers to prepare special equipment – they only require a mobile phone.

The following benefits arise from such a solution:

- the system will be low-cost, which makes it convenient to deploy it in an educational setting;
- it will increase usability because people can capture data by simply using their own devices.
- real-time video allows for real-time processing – that means object detection, recommendation, and robot actions.

Thus, proposed communication architecture will ensure perfect synchronization of perception, decision-making, and action components of the robotic pipeline.

3.2. Commercialization Aspect of the Product

The proposed system is created with a well-thought-out commercialization path within the field of assistive educational robotics for specially needed children, being created together with SERD (Services for Empowerment and Rehabilitation of Disabled Children and families). The company SERD is engaged in providing an environment for the children who have any needs to develop their mental abilities in a structured form, combining emotional care, education, and teaching skills. In such a case, the system can be implemented in classroom conditions as a tool for helping to improve cognitive and motor skills by means of robotic object sorting exercises.

As far as commercialization opportunities are concerned, the system can become a part of educational robotics learning kits that will unite computer vision technology, predictive machine learning, explainable artificial intelligence, and robotic actuation into one

platform. The core selling point of the kit would be the capability of turning abstract notions of classification, grouping, comparing, and making decisions into something tangible and interactive. By using three different shapes and three colors for defining the objects, one will make them easily recognizable for the children.

The sorting and system work take place in an environment where the child learns in the presence of the guidance of the tutor. In this instance, the tutor introduces the idea of sorting and the system's operations, places objects on the working area, and starts the operations using a basic control operation. The robot performs a predictive color-based sorting mode, where it sorts objects according to the predominant color into their respective bins. Through this kind of continuous engagement, the system becomes a means to demonstrate how sorting works many times prior to the learners carrying out the activity alone.

One of the major advantages of this system is that it is able to work in two modes, where learners learn through demonstration and later engage in the process themselves. The system can be utilized to watch a demonstration whereby the robot sorts through objects a number of times and then the learner imitates by manually sorting objects placed on the surface. Through this method of learning, learners get involved directly in their education process.

The integration of Explainable Artificial Intelligence (XAI) enhances greatly the market value of this product within educational institutions. The system is not only capable of performing sorting operations but also explaining the reasoning behind certain decisions, such as choosing a specific color due to distribution of objects in a scene and inference of a model. The system allows teachers to understand how it works and explain the principles of operation to the students.

As far as technical characteristics of the system go, they include being low-cost, modular, and scalable enough for implementation in limited-resource learning facilities, such as SERD-affiliated centers. The use of readily available hardware parts, the ability to stream from a smartphone-based IP camera, and the use of microcontrollers to perform actions make sure the system will not need any costly industrial equipment.

Commercialization-wise, this system is more focused on making use of the latest technologies like object detection using YOLO, predictive sorting using machine learning, and explainable AI to create an effective educational system. This system combines robotics technology and inclusive education as it offers an interactive approach for better learning and efficient teacher-led instruction. This makes it easier for special kids to participate in constructive cognitive development tasks using robots.

3.3. Testing and Evaluation

The testing process will be performed on the complete implementation of the proposed system. The testing process aims at assessing the validity and consistency of the integrated systems components, such as object recognition, prediction of recommendations, robotic operations, and explainability of results produced by artificial intelligence. Given that the system will be used for an educational purpose involving specially-needed children in collaboration with SERD, the testing process also includes the analysis of whether the system can operate uniformly in a controlled setting.

To test the operation of the system, the first step to be made is to check whether the vision system is functioning effectively. To this end, the YOLOv8 algorithm should be analyzed based on the capability of correctly detecting and recognizing objects in the video stream. Testing will be done in terms of various light conditions, object positions, as well as

varying numbers of objects. All recognized objects will be compared to their correct labeling in order to check whether their class corresponds to any of the nine pre-defined categories.

Next comes the testing of the predictive recommendation module, where the machine learning models developed previously create the sorting strategy using extracted features, such as the number of objects, colors, and shapes found. The accuracy of the recommendation made will be evaluated using the comparison between the predicted and the expected outcome, in terms of how the objects should be sorted according to the dominant color.

Lastly, the robotic execution module will be tested to confirm if the robot arm controlled by the Arduino correctly picks and places the objects based on the commands it receives. Every movement task will be individually checked in order to determine if the servos move accurately and smoothly, and if the location is properly detected based on the grids for objects. It is especially important to check if there are no communication problems between the Arduino and the Python program.

Meanwhile, the Explainable AI module will undergo the tests for transparency and understandability in order to prove whether the generated explanations accurately convey the underlying logic of system behavior. The outputs of feature contribution based on SHAP will be confirmed to verify that the most critical attributes match the system decisions. Thus, explanations produced by the Explainable AI module should be consistent in their content depending on the scenario, enabling users to understand and interpret the actions of a robot.

Lastly, a joint test for the system will be conducted, when all modules work in tandem within the process. In particular, the sequence from object detection through predictions, explanations generation, and robotic actions is going to be continuously evaluated. Sorting

efficiency, speed, and responsiveness of explanation in such a complex task will be assessed in the end-to-end performance of the system. Also, regarding the educational purpose of SERD applications, observational test will be held in order to evaluate interaction between users and the system. The test phase is aimed at ensuring that the system is not only functional but also robust and able to provide explanations in a reliable manner.

4. Results and Discussion

An analysis of the suggested predictive robot sorting framework was done in an experimental educational setting which catered to students with different abilities from SERD Kelaniya. The system involves computer vision, intelligent decision-making by means of machine learning, explainable AI and robot operation for recognition and sorting of objects through the use of color learning techniques.

4.1. Results

4.1.1. Grid Mapping and Spatial Localization

Homography successfully mapped camera coordinates to grid coordinates through perspective mapping, thus achieving:

- Identification of square ID where objects were located
- Mapping of detected objects to robotic pick points
- Spatial consistency despite varying camera perspectives

Therefore, any detected object could be both classified and localized in the grid.

4.1.2. Object Detection system

The object detection performance of the YOLOv8 model trained for 50 epochs was evaluated using standard evaluation metrics including Precision, Recall, and Mean Average Precision (mAP).

Table 4. 1 Performance Evaluation of YOLOv8 Object Detection Model:

Metric	**Value**
Precision	0.949
Recall	0.956
mAP@50	0.954
mAP@50–95	0.543

The model produced a Precision of 0.949, meaning that around 94.9% of the objects identified by the model are true positives and there are no false positives, which is a strong indicator that the model does not generate a large number of false positives. The Recall of 0.956 suggests that the model can identify 95.6% of all objects in the dataset correctly. This signifies that the model has very few false negatives, which is a vital quality for a robotic sorting algorithm that needs to detect every object before picking and sorting.

mAP@50 of 0.954 denotes that the overall performance of the model in detecting and classifying objects in a given test set, assuming an IoU threshold of 0.5, is satisfactory.

Nevertheless, the mAP@50–95 of 0.543 displays a significant decline in performance when using higher IoU values. It is evident that although the model exhibits outstanding

performance in object detection, it falls behind when it comes to more accurate localization. Such observations are frequent when applying models to practical robotic vision applications, particularly when objects differ in terms of position, illumination, and obstruction.

Overall, the YOLOv8 model exhibits great performance when detecting and classifying objects, making it a perfect choice to serve as the perception component in the designed robotic system. Figure 4.1 shows how objects were successfully identified on a tricky background surrounded with similar shape objects, avoiding them.

Figure 4. 1: Object Detection output

4.1.3. Machine Learning Model Performance

4.1.3.1. Performance Evaluation of Recommendation Models

The machine learning module was designed to determine the most efficient sorting strategy considering the distribution of objects in terms of their color, shape, and count characteristics. The following classifiers were considered for the problem: Random Forest, XGBoost, and Logistic Regression.:

Table 4. 2: Performance Comparison of Recommendation Models

Model	**Accuracy**
Random Forest	0.94
XGBoost	0.88
Logistic Regression	0.88

The observations from the experiment show that Random Forest had the best accuracy (0.94), followed by XGBoost and Logistic Regression, whose performances were similar. This means that the data set has some overlapped decision boundaries because the three models worked almost equally well on the data set.

4.1.3.2. Random Forest Detailed Evaluation

Given that Random Forest was chosen as the final machine learning model for the development of the recommendation system, its effectiveness was assessed through other classification metrics as well.

Table 4. 3: Random Forest Detailed Evaluation

Metric	**Value**
Accuracy	0.94
Precision	0.93
Recall	0.74
F1-score	0.82

As can be seen from the results, this classifier demonstrates excellent precision, equal to 0.93, which means that the probability of the predicted instance being correctly categorized into a particular class is extremely high. In the context of an educational robotic system, incorrect classification might cause discrepancies in the feedback delivered.

At the same time, recall shows that this algorithm cannot equally recognize various strategies, which explains why this indicator only equals 0.74. In other words, while being rather reliable in predicting certain instances, the Random Forest classifier fails to identify others, which is explained by their smaller amount in the database.

Nevertheless, the F1 score equals 0.82, which proves that the balance between precision and recall is maintained successfully.

4.1.3.3. Model Selection and Justification

While XGBoost and Logistic Regression demonstrated roughly equal results regarding accuracy, Random Forest was ultimately chosen as the recommended model because of performance consistency and additional requirements at the system level.

One of the important reasons for this choice was related to integration with the Explainable AI (XAI) component, where interpretability became critical. Random Forest works very efficiently when applied in combination with SHAP explanations to identify how input features affect the recommendation outcome.

Random Forest gives us the following advantages compared to other models:

- Stable predictions for different combinations of objects
- Good support for feature importances interpretation
- More insightful explanations for users (teachers, students)

With respect to this study, the aspect of interpretability was regarded as equally important as that of predictive accuracy because it was critical for the system's educational application, which requires that explanations be given for each sorting decision taken. It was for this reason that Random Forest emerged as the ultimate recommendation engine model due to its superior level of interpretability besides being the most accurate algorithm.

4.1.4. System-Level Sorting Performance

For the testing of the entire system, some objects were positioned randomly across the grid, and the sorting procedure was conducted in real-time. This sorting starts with the object detection via YOLOv8, and is followed by grid localization and classification in which the detected objects are located within a specific area. Once all objects have been localized, the machine learning algorithm creates a sorting plan based on the object location and focuses primarily on color classification. The sorting plan is subsequently translated into actions that are sent to the Arduino.

Overall, the test results showed the successful operation of the full system in which all objects that had been positioned within the designated grid area were detected in real-time, and a sorting order based on the previously determined strategy was created by the machine learning algorithm. All required actions were successfully performed by the robotic arm without any collisions or errors. Therefore, the interaction between all three layers – perception, reasoning, and actuation proved effective for the sorting tasks assigned.

4.1.5. Arduino Robotic Execution

Robotic control system based on Arduino effectively converted higher-level commands to specific motions. Key Observations are,

- Smooth servo motion minimized jerks through incremental motion control
- PCA9685 PWM chip controlled servos in synchronization
- LM2596 chip provided constant power output to motors
- Programmed motion sequences (pickBox1-pickBox22) facilitated control

Successfully performed tasks were:

- Picking up objects from 22 locations
- Placing picked objects in three categories (Red, Blue, Green)
- Returning to the initial position continuously

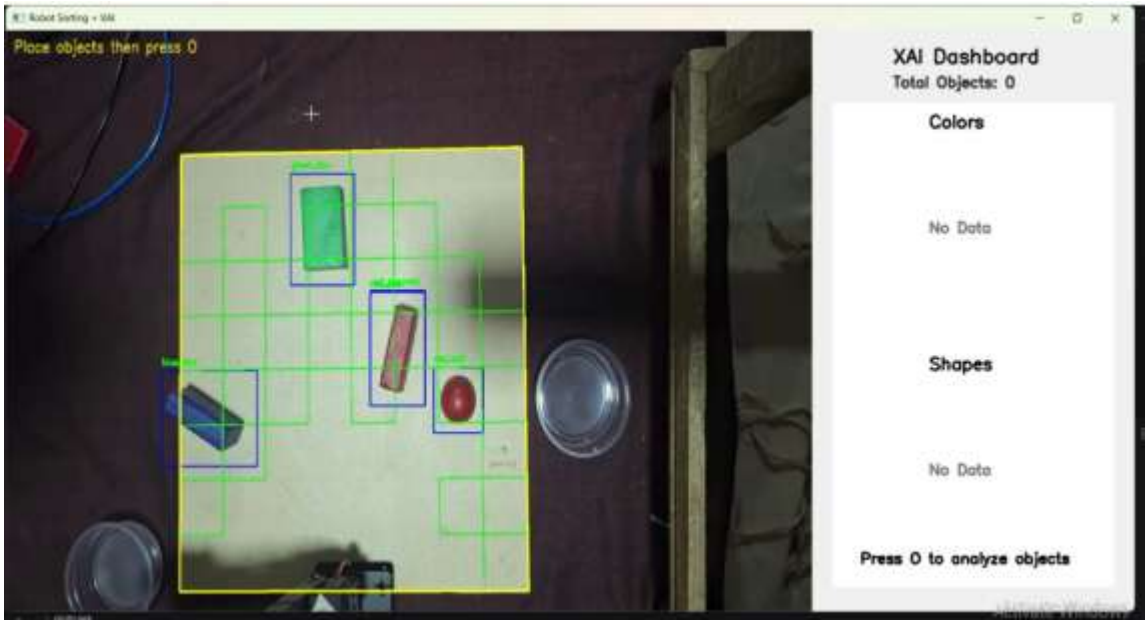

Figure 4. 2: Successfully detected objects and waiting for command

Python-Arduino serial connection allowed immediate task performance without any delays.

4.1.6. Explainable AI (XAI)

In order to increase the level of interpretability, the proposed approach is integrated with an Explainable AI (XAI) block based on reasoning approaches inspired by SHAP methods. This part was developed particularly in order to assist teachers and non-professional users in interpreting the recommendations provided.

The visualization of object distribution includes frequency graphs for colors and shapes which demonstrate the structure of the detected objects in a specific period. In addition to that, there is an explanation panel which provides information about the rationale used to come up with the sorting recommendations. Such explanations include simplified textual representation like the one provided for the Random Forest model used for sorting.

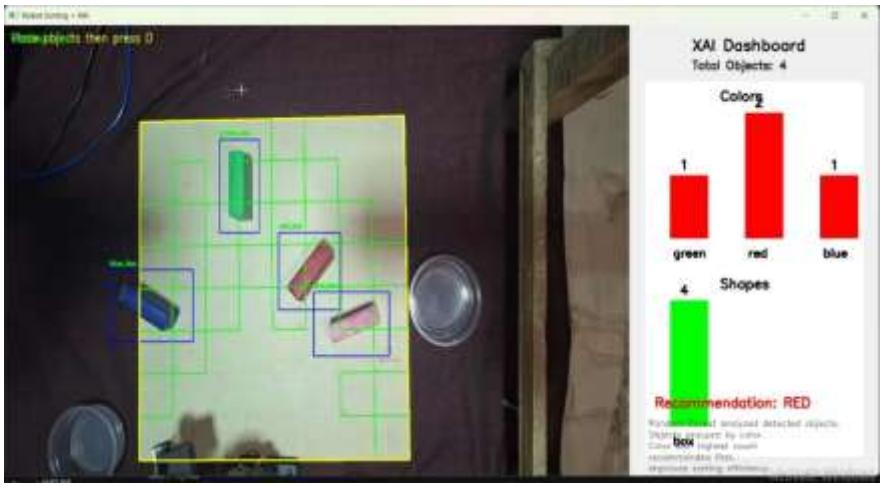

Figure 4. 3: XAI output while sorting is happening

This additional layer of interpretation allows not only seeing the outcome of the process but also figuring out why such and not other decision has been made. The educators will have the opportunity to observe the effect of various features of the objects on the recommendation and the distribution patterns in respect to their importance for the prioritizing process.

4.2. Research Findings

Based on the carried-out study, it is possible to state that by combining computer vision, machine learning, explainable AI, and robotic control, it is possible to achieve a rather efficient predictive sorting system that could be used as an assistive tool in educational settings. This research proves that by applying object detection in combination with spatial mapping it is possible to accurately identify and localize various objects that have been randomly placed on a physical grid.

Firstly, according to the results of the experiment, YOLOv8 shows high performance in the process of real-time object detection, providing high precision and recall. This is why it is considered a viable choice for creating a perceptual layer in robotic learning platforms.

Moreover, based on the obtained results, it could be stated that machine learning algorithms could significantly contribute to improving the decision-making process in comparison with rule-based approaches. By comparing Random Forest, XGBoost, and Logistic Regression models, it is possible to conclude that the first two approaches allow obtaining better prediction results in terms of stability and accuracy of strategy recommendations.

The research also demonstrates the importance of interpretability in AI decisions for education-related applications. The use of XAI technology allows making processes of making predictions more transparent by giving an opportunity to understand what lies behind the decision. Feature explanations and visualized object distribution help in bridging the gap between AI decisions and humans' perception.

Additionally, the research proves that robotic actions can be performed online due to the interaction between intelligence built in the Python system and robot built in the Arduino system. This interaction helps coordinate all actions and execute pick and place activities in different grids. In conclusion, it should be noted that the use of prediction models together with explainability and robotic actions results in creation of a structured educational process where students can acquire skills in object recognition and classification understanding.

4.3. Discussion

The results of this research prove that a well-coordinated system based on computer vision, machine learning, explainable artificial intelligence, and robotics could provide effective assistance in real-time predictive sorting tasks. The main aspects that will be discussed in the following part are the contribution of each module, the relations between them, and the practical implications of the obtained results for assistive education.

The first layer – object detection using YOLOv8 – is crucial for achieving the highest accuracy of environment perception. The high accuracy rate of both precision and recall proves that this algorithm is able to accurately recognize and classify objects regardless of various conditions. It is important since even a small mistake at this stage leads to wrong predictions and actions in the subsequent stages.

The recommender layer based on machine learning elevates the level of intelligence through a step further away from sorting based on rules. When assessing the performance of the Random Forest classifier, XGBoost, and Logistic Regression algorithms, we can conclude that the problem requires an ensemble method, which is generally better adapted

for making decisions based on a complex combination of multiple parameters. The difference in performance is insignificant, but, nevertheless, Random Forest offers a good compromise between accuracy and consistency, especially in terms of handling interacting features, e.g., the shape and color distribution.

An important point about the developed solution is that it incorporates Explainable AI into the framework in order to solve the interpretability problem associated with machine learning applications. SHAP inspired explanations allow us to understand the decision process in terms of features' contribution and object distribution. The ability to visualize the decision process is vital in the field of education as it builds trust in the system and contributes to the achievement of learning goals.

The robot execution layer proves that physical engagement can be performed consistently through well-defined motion planning and serial communication between the software and hardware modules. The employment of the pre-defined motion patterns and smooth servo transitions makes the robot behavior consistent. Nevertheless, the efficiency of the processes is affected by mechanical tuning and the physical environment setup, which means that physical precision is still a limitation compared to the software part.

Generally, the presented system represents an example of high interdisciplinary consistency as it unites several functions in a chain: perception, reasoning, explanation, and actions. What is the key advantage of the system in question? It consists in performing all the required processes simultaneously and providing interactive learning. At the same time, the system reveals the problem of model generalization and precision limitations in physical space.

5. Conclusion

Thus, the project was able to create a working robotic sorting system using computer vision, machine learning, explainable AI, and robotic control techniques in a single coherent framework. Specifically, the proposed system implements a chain of intelligent processes starting with object detection with YOLOv8, object location using grid-based mapping, decision-making with machine learning models, and robot manipulation by controlling an Arduino-driven robotic arm. It results in the formation of an efficient mechanism that connects software intelligence with hardware actuation.

With regard to data science, one can note the successful use of different machine learning algorithms for decision-making problems. In particular, the comparison of the Random Forest, XGBoost, and Logistic Regression models reveals higher performance and robustness of models based on ensemble techniques when processing structural information related to the color, location, and other characteristics of objects. The implementation of such preprocessing methods as data encoding and rebalancing also contributed to model improvement. Among all the implemented algorithms, the Random Forest model can be regarded as the best choice for practical applications.

One of the primary contributions of this project is the inclusion of the Explainable AI component that turns the system into a comprehensible learner rather than a simple automated decision-making agent. The possibility of receiving visual and written explanations of decisions made by the machine enables people to comprehend how their choices are made as opposed to simply seeing the results.

Apart from technical aspects, the significance of the project is also reflected in its pedagogical and societal implications. Although there are many other similar projects that implement the use of intelligent sorting or robotic systems, these are usually aimed at generalized purposes rather than specific cases. One of the main advantages of this particular system is that it helps to visualize certain abstract concepts associated with classification, comparisons, and identification through physical interaction with objects and assistance provided by robots.

The importance of this research context is even greater when it is realized that such AI and robot technologies are usually built and utilized in typical environments created for ordinary people who do not require any additional education or adaptation. At the same time, the aim of the current system is concentrated on the less investigated aspect, as the technology itself has been altered to address the needs of students with varying abilities. The change in the sphere of utilization is aimed at showing a more socially oriented perspective on intelligent technologies, which involves not only high effectiveness but also access, flexibility, and interaction with various people.

In terms of introducing the system to assistive education, this study proves that modern technologies of data science and robotics can create even greater value for society when applied for underdeveloped areas. In this regard, the unique contribution of this research consists of the fact that the technology used is not only automated and optimized for industrial production but also contributes to cognitive and learning processes.

As a result, this project signifies a move away from design that focuses on technological components to human-focused and inclusive design. This demonstrates that when machine learning, explainability, and robotics are integrated with the education process, the end result becomes much more than a mere functionality—it becomes something socially significant as well.

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